ANIMESH SINGH

Versatile software expert delivering full-cycle application development, transforming complex business needs into scalable solutions.

Mobile: +91 9179953607 **Email:** animeshs34@gmail.com

Profile Summary

Dynamic & Results-Oriented **Software Engineer with 4+ years of experience** in backend engineering, database administrations, system integration,cloud-native solutions, and Generative AI applications. Adept at designing scalable, high-performance systems across diverse tech stacks and quickly mastering emerging technologies. Proven track record of leading complex projects, optimizing performance, and **delivering innovative solutions** in fast-paced environments.

- Expertise in building robust applications using **Golang**, Python, PostgreSQL, and **microservices architecture**, with strong software design and best practices.
- Hands-on experience in Generative AI (LLM integration, LangGraph, **RAG pipelines**) for campaign automation, lead personalization, and **AI-driven enterprise features.**
- Skilled in cloud-native development (AWS, GCP), API integrations, and security protocols (**OAuth, SAML, SSO**) ensuring secure and seamless interoperability.

Core Competencies and Technical Skills

Backend Development
Golang
Python
Java

Database Management
PostgreSQL,Oracle RDBMS
SQL, Migrations

Cloud & DevOps
GCP (Google Cloud Pub/Sub)

API & Authentication
OAuth, SAML, SSO, Google & Okta Authentication

Payment Processing Stripe Integration, Razorpay Integration

System Optimization – Debugging, Troubleshooting, Root Cause Analysis

Generative & Applied AI
AI/ML with TensorFlow,Neural Networks (LSTM, Autoencoders)
LangGraph, RAG Pipelines, Embeddings, Vector Databases

Microservices Architecture and Event-Driven Systems
NATS Streaming, gRPC, RESTful APIs, Kafka, RabbitMQ, Google Pub/Sub
Docker, Kubernetes, Helm, Distributed Tracing (Prometheus, Otel), Fault-Tolerant Design
AI/ML Service Integration

Work Experience

Software Engineer

Appointy Information Technology Pvt. Ltd., Bhopal
Aug'21- Till Now

Technologies and Integration Used: Google Calendar integration, Okta and Google SSO , Stripe Integration, GCP pub/sub, PostgreSQL, NATS Streaming Server,LangGraph, LLMs, RAG Pipelines, Vector Databases (pgvector, Pinecone), Machine Learning algorithms , Core Neural Networks (AutoEncoders and LSTM NN), TensorFlow

Key Result Areas:

- Spearheading the project lifecycle: feasibility through scoping, requirements definition through specification, design, development, quality assurance, integration and implementation
- Managing client /user interaction for requirement gathering, risk assessment, finalization of technical architecture; holding discussions for effort/ cost/ time estimation and reporting the same
- Administering service delivery functions, to ensure end-to-end ownership of incidents and service requests

- Meeting deadlines and Turn-Around-Time without compromising quality norms and adhering to SLA
- Handling new and existing features in ongoing enterprise project Google REWS Scheduling software
- Integrating Okta and Google SSO authentication with OAuth and SAML specification
- Working in a team for developing microservices intercommunication using NATS streaming server
- Supporting bug fixing, maintaining store procedures and executing system enhancements according to user requirements for the application
- Built a self-hosted Identity Provider using Ory Hydra & Kratos, enabling OAuth2, OpenID Connect, and user authentication
- Designed and deployed a GenAI-based lead personalization and campaign automation system, leveraging LLMs, LangGraph, embeddings, and RAG pipelines to generate hyper-personalized B2B outreach emails.
- Applied machine learning models for intelligent scheduling recommendations and deep learning-based NLP pipelines to enhance contextual understanding in customer interactions.
- Built vector search systems using pgvector for fast semantic retrieval in enterprise applications.
- Deployed dockerized local LLM environments ensuring cost-efficiency, data privacy, and scalable inference without dependency on external APIs.
- Enhanced ongoing enterprise Scheduling Software with AI-driven automation and predictive analytics.

Highlights:

- Improved overall system load time(HTTP request times of various request significantly reduces by 60-70%(independent of size of data
- Developed payment gateway to handle one time as well as recurring payment globally through stripe payment gateway integration
- Contributed to AI-powered automation features that reduced manual workflows and boosted productivity.

Internship

Data Analytics and Python Intern
Appin Technology Lab , Bhopal(M.P.)
Jul'19- Dec'19

Key Role:

- Gained practical knowledge of Python basics and applied it to design a Web Crawler which efficiently fetched valuable information from various web applications and processed it to extract useful data
- Explored machine learning fundamentals and various algorithms to train statistical models, used Neural Networks to predict images and improve edge detection between multiple subjects, achieving an accuracy of up to 81.24%
- Acquired a foundational understanding of big data technologies such as Hadoop, Hive warehouse, Hive Query Language, and map-reduce

Education

BE+ M. Tech, Integrated PG Programme (Computer Science)
Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal
2021; scored 8.13 CGPA

12th
Kendriya Vidyalaya, Sehore (MP)
2016; scored 86%

10th
Kendriya Vidyalaya, Sehore (MP)
2014; scored 8.6 CGPA

Academic Projects

Minor Project, 2018
University Unified Students Portal

Technology Used: Java enterprise(SERVLETS and JSP) and MySQL 5.1 database and Ajax for more interactive UX

- Created a fully functional and deployable portal that consolidated all student details in one centralized location, providing them easy access to their information

Java Desktop Application, 2019

Restaurant billing system

Technology Used: Java Swing, Oracle 10 Database

- Developed desktop based application to handle all possible complex operations performed in real work billing counter machines

Major Project, Dec,2020

Location Specific Future AQI Predictor

Technology Used: Machine learning and Neural Networks(auto-encoder neural networks with ensemble of support vector machine)

- Analyzed real world air quality index(AQI) data for India and successfully predicted future AQI's over time at an efficiency of 91.56%

Research Paper

- Published Research Paper on "A Survey On Deep Learning Approaches For Development More Robust And Reliable IDS Models" in 2021
  http://thedesignengineering.com/index.php/DE/article/view/5191

Personal Details

Date of Birth	: 02/02/1999
Languages Known	: English & Hindi
Address	: Bhopal